RAJALAKSHMI ENGINEERING COLLEGE

Barath S 251801040-2025-AIDS B2

Started on	Thursday, 27 August 2026, 11:30 AM
State	Finished
Completed on	Thursday, 27 August 2026, 11:32 AM
Time taken	1 min 25 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Problem Statement:

Given a sorted array of integers say arr[] and a number x. Write a recursive program using divide and conquer strategy to check if there exist two elements in the array whose sum = x. If there exist such two elements then return the numbers, otherwise print as "No".

Note: Write a Divide and Conquer Solution

Input Format

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Sum Value

Output Format

First Line Contains Integer – Element1

Second Line Contains Integer – Element2 (Element 1 and Elements 2 together sums to value "x")

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>
int binarySearch(int* arr, int low, int high, int target) {
    if (low > high) {
        return 0;
    }
    int mid = low + (high - low) / 2;
    if (arr[mid] == target) {
        return 1;
    } else if (arr[mid] > target) {
        return binarySearch(arr, low, mid - 1, target);
    } else {
        return binarySearch(arr, mid + 1, high, target);
    }
}
int findPair(int* arr, int low, int high, int x, int* num1, int* num2) {
    if (low >= high) {
        return 0;
    }
    int current = arr[low];
    int target = x - current;
    if (binarySearch(arr, low + 1, high, target)) {
        *num1 = current;
        *num2 = target;
        return 1;
    }
    return findPair(arr, low + 1, high, x, num1, num2);
}
int main() {
    int n;
    if (scanf("%d", &n) != 1) return 0;
    int arr[n];
    for (int i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
    }
    int x;
    scanf("%d", &x);
    int num1 = 0, num2 = 0;
    int found = findPair(arr, 0, n - 1, x, &num1, &num2);
    if (found) {
        printf("%d\n", num1);
        printf("%d\n", num2);
    } else {
        printf("No\n");
    }
    return 0;
}
```

	Input	Expected	Got	
✔	4 2 4 8 10 14	4 10	4 10	✔
✔	5 2 4 6 8 10 100	No	No	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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